

Owusu Kantankah

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EDUCATION

University of Wisconsin–Madison

Madison, WI

Bachelor of Science in Computer Science and Data Science

Expected May 2028

- **Coursework:** High Performance Computing (Graduate), Computer Graphics, Algorithms, Databases

EXPERIENCE

LASER Research Program, UW–Madison

Madison, WI

Software Engineer Intern

May 2026 – Aug. 2026

- Built an end-to-end PyTorch/CUDA multi-agent reinforcement-learning system training **100 concurrent agents** across three interchangeable network architectures for controlled architecture experiments.
- Improved training throughput **2.3x** ($0.884s \rightarrow 0.384s$ per episode) by batching per-group linear layers with `torch.bmm` and reducing GPU launch overhead using `torch.compile` and CUDA graph capture.
- Deployed trained policies to a real-time Three.js client with TensorFlow.js by reimplementing the network forward pass in JavaScript and rendering **100 instanced agents**.
- Received the L&S Summer of Excellence in Research **Rising Researcher Award**; earned a return offer to continue research through the academic year and was selected as a Spring 2027 Computer Graphics TA.

Space Science and Engineering Center (SSEC)

Madison, WI

Undergraduate Data Engineering Researcher

Sep. 2025 – May 2026

- Built a Python pipeline with NumPy and pandas to process **10GB+** of NetCDF telemetry from JAXA's Akatsuki orbiter, reducing analysis runtime by **35%**.
- Analyzed 1440×720 atmospheric grids to identify a ~ 195 K polar vortex and 20 K equator-to-pole thermal gradient; presented findings at UW–Madison's Undergraduate Research Symposium.

StudioNil

Remote

Founder & Software Engineer

Mar. 2023 – Aug. 2025

- Founded and engineered a multiplayer game that reached **10,000 monthly active users in six weeks**.
- Architected client-server infrastructure supporting **100 concurrent players per instance** while maintaining a **0.02% crash rate**.
- Built a custom 3D spatial hitbox system using raycasting, improving collision accuracy by **1.8x** over Roblox's native .Touched API.

PROJECTS

Godot Game Engine (Open Source) | C++, SCons, Git | [GitHub](#)

- Authored an upstream-merged fix for a Godot 4.7 animation regression by reproducing it across releases, using `git bisect` to isolate the offending change, and tracing it through the C++ animation pipeline.
- Corrected a non-commutative matrix-composition error ($AB \neq BA$) in 3D transform scaling, applying the correct local-space basis operation to preserve non-uniform scale during TRS animation.

3D Drone Simulation | Three.js, GLSL, JavaScript, HTML/CSS | [GitHub](#)

- Built a 3D drone-swarm simulation implementing Reynolds' boids algorithm across **1,000 agents**, optimized for real-time performance through instanced rendering and spatial hashing.
- Collapsed **1,000 draw calls** into one `InstancedMesh` dispatch, improving frame rate from **80 to 133 FPS** and reducing GPU memory usage to **9 MB**.
- Replaced $O(n^2)$ pairwise neighbor search with a uniform spatial hash grid using a 27-cell lookup, reducing neighbor queries to near-linear time while keeping the physics loop allocation-free.

Centralized Morgridge Event Calendar | Python, Flask, SQLite, AWS | [GitHub](#)

- Built a RESTful Flask/SQLite API for event retrieval and channel management on a **500-member** Discord server, deployed on AWS EC2 as the data layer between automated scrapers and a React frontend.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, GLSL, HTML/CSS

Frameworks/Libraries: PyTorch, TensorFlow.js, React, Flask, Three.js, NumPy, pandas, Tailwind

Tools: CUDA, Docker, AWS (EC2), Linux, CMake, Git, SQLite